Novia One.

FMP · MODULE 4 · WORKBOOK

PRACTICAL EVIDENCE · EQUITY SECURITIES

Workbook, answered.

Model answers to every question in the Module 4 workbook — all **10 questions, 150 marks** — written to match the lecturer's marking style: **bullet points with examples** for list questions and **prose** for conceptual ones, comprehensive throughout.

Note: these are study/model answers to learn from. The workbook is assessed as **your own work** (pass mark **65%**) — reword them in your own voice for your submission.

MARK ALLOCATION AT A GLANCE

QUESTION	TOPIC	MARKS
Q1	Equity fundamentals & capital structure	15
Q2	Equity income, share types, listed/unlisted, DRs	15
Q3	Determinants of optimal capital structure	15
Q4	Debt versus equity	10
Q5	Primary & secondary markets, IPOs, listing	30
Q6	Listing reasons, methods, rights issue, risks	15
Q7	Gordon Growth model — shortcomings	5
Q8	Market capitalisation (FSR vs Nedbank)	10
Q9	Valuation techniques by sector	10
Q10	Dual-class shareholding case study	25
Total		**150**

ANSWER STYLE List-type questions use a lead-in line + bold-label bullets (with examples); conceptual "why/how/difference" questions are answered in prose. Each answer gives more points than the mark count, and both sides where relevant.

QUESTION 1 · EQUITY FUNDAMENTALS & CAPITAL STRUCTURE [15]

1.1 Main advantages of raising equity capital for a company? (3)

The main advantages of raising equity capital include no additional financial burden from interest payments, no obligation to repay the capital, no mandatory fixed payments (dividends are discretionary), and no maturity date — all of which give the company valuable operational and financial flexibility. In addition, bringing in large or institutional investors can provide business expertise, resources and valuable contacts, while a stronger equity base lowers gearing and supports future borrowing.

1.2 Factors that influence the optimal capital structure? (3)

The optimal capital structure of a company is influenced by factors such as the corporate lifecycle, the tax-deductibility of interest, business risk, bankruptcy (financial-distress) risk, agency costs, and lender risk appetite. Together these factors determine the ideal balance of debt and equity — the mix that minimises the company's cost of capital and maximises its market value.

1.3 Purpose of equity in the capital markets? (3)

The purpose of equity in the capital markets includes raising capital for businesses through share offerings, providing a marketplace for buying and selling shares, protecting investors against fraud, ensuring regulatory compliance, providing real-time price information, maintaining an orderly market, and offering investors a share in company profits.

1.4 Role of equity in a company's capital structure? (3)

Equity in a company's capital structure represents ownership in the business through common and preferred stock, together with retained earnings. It offers advantages such as no obligation for repayment or interest payments and greater operational flexibility, but it can also lead to a loss of control and to profit-sharing with new shareholders through dilution.

1.5 How does the Debt-to-Equity (D/E) ratio help assess financial risk? (3)

The Debt-to-Equity (D/E) ratio measures the financial risk of a company by comparing its total debt to the value of its shareholders' equity. A higher D/E ratio indicates greater reliance on borrowed funds and therefore greater financial risk — more fixed obligations and higher exposure to default in a downturn — whereas a lower ratio indicates a more conservatively financed, lower-risk company. For example, a D/E ratio of 2.5 means the company has R2.50 of debt for every R1.00 of equity.

QUESTION 2 · EQUITY INCOME, SHARE TYPES, LISTED/UNLISTED, DRS [15]

2.1 Two primary forms of income from investing in equities? (3)

The two primary forms of income generated from investing in equities are:

- **Capital growth:** the increase in the value of shares when their market price rises — for example, if a share bought for R100 rises to R110, the investor gains 10% in capital growth.
- **Dividends:** payments made by a company to its shareholders out of its profits, typically paid on a regular basis, offering a steady income stream. (For foreign shares, foreign-exchange gains or losses can be a further source.)

2.2 How does investing in preference shares differ from ordinary shares? (3)

Investing in preference shares differs from investing in ordinary shares in several ways:

- Preference shareholders receive their dividends **before** ordinary shareholders and typically have a **fixed** dividend rate.
- Ordinary shareholders have **voting rights** and participate in company governance, whereas preference shareholders usually do **not** have voting rights unless specified.
- In the event of **liquidation**, preference shareholders have **priority** over ordinary shareholders in receiving their share of the company's assets.

2.3 Significance of voting rights for ordinary shareholders? (3)

Voting rights for shareholders in ordinary shares are significant because they allow investors to participate in key decisions within the company. This includes voting on issues such as the election of directors, mergers, and the appointment of auditors. Shareholders can exercise their voting rights at the annual general meeting, either in person or via proxy. These voting rights provide shareholders with influence over the company's management and strategic direction.

2.4 Main differences between listed and unlisted equity securities? (3)

Listed and unlisted equity securities differ in several ways:

- **Market & trading:** listed shares trade on a licensed exchange (a public market) with continuous trading, while unlisted shares trade over-the-counter through dealers or market makers (a private market) with irregular trading.
- **Regulation & transparency:** listed shares are regulated under the Financial Markets Act with transparent pricing; unlisted shares have little regulation and opaque pricing.
- **Liquidity & risk:** listed shares are liquid and lower-risk, whereas unlisted shares are illiquid, carry wide bid-ask spreads and the potential for price manipulation, making them riskier.

2.5 Role & characteristics of a depository receipt (DR)? (3)

A depository receipt (DR) is a negotiable instrument issued by a bank that represents an interest in a foreign company and trades on a domestic exchange just like a common share. Its role and characteristics are:

- **Role:** it facilitates cross-border investment at a lower cost, in the investor's own market and currency.
- **Issued by a bank, not the company:** so it raises no capital for the underlying company.
- **No voting rights:** these rest with the depository bank, not the investor.
- **Forms:** known as a GDR (Global Depository Receipt) globally, and an ADR (American Depository Receipt) in the United States.

QUESTION 3 · DETERMINANTS OF OPTIMAL CAPITAL STRUCTURE [15]

3.1 How does a company's lifecycle stage influence its optimal capital structure? (3)

A company's lifecycle stage strongly influences its optimal capital structure. Start-up and growth companies have uncertain cash flows and few tangible assets, so they rely mainly on equity such as venture capital and carry little debt. Mature companies, with stable cash flows and a solid asset base, can support more debt and benefit from the interest tax shield. Companies in decline tend to reduce debt or return capital, since over-gearing at this stage sharply raises the risk of financial distress.

3.2 Why is business risk a determinant of optimal capital structure? (3)

Business risk — the variability of a company's operating earnings — is a key determinant because taking on debt layers financial risk on top of it. A firm with high business risk that also carries high debt compounds its total risk and raises the probability of financial distress. Such firms should therefore use less debt to keep total risk manageable, whereas firms with stable, predictable earnings can safely carry more.

3.3 Why does increasing leverage initially raise firm value, and what risk offsets this? (3)

Increasing leverage initially raises firm value because debt is cheaper than equity and interest is tax-deductible; this tax shield lowers the weighted average cost of capital and lifts the firm's value. The offsetting risk is financial-distress or bankruptcy risk: as debt rises, so does the probability and cost of distress. Beyond the optimal point these distress costs outweigh the tax-shield benefit and firm value begins to fall — the essence of trade-off theory.

3.4 What role does lender risk-appetite play for early-stage companies? (3)

Lender risk appetite is critical for early-stage companies because they are inherently risky, with no track record, few assets and uncertain cash flows. Lenders are therefore reluctant to advance funds and demand high interest rates or collateral that the company cannot provide. This limited willingness to lend effectively caps the amount of debt available, forcing early-stage firms to rely on equity such as angel or venture-capital funding.

3.5 How do agency costs affect capital-structure decisions? (3)

Agency costs affect capital-structure decisions because of conflicts between the parties involved. The agency cost of equity arises when managers misuse free cash flow, for example through empire-building; taking on debt imposes discipline by committing that cash to fixed repayments. The agency cost of debt arises when shareholders take excessive risk or underinvest at the lenders' expense, which prompts lenders to impose covenants and charge higher rates. The capital structure is set to balance these opposing costs.

QUESTION 4 · DEBT VERSUS EQUITY [10]

4.1 Primary difference between debt and equity (obligations & ownership)? (2)

The primary difference is that debt securities represent a liability for the issuing company, which must repay the principal and pay interest at specified intervals. In contrast, equity securities do not impose a repayment obligation on the company; instead, equity shareholders are owners with a residual claim on the company's assets after all liabilities have been paid. Equity investors are not guaranteed periodic payments, but they seek a total return from capital appreciation and dividends.

4.2 How do equity investors' objectives differ from debt investors'? (2)

Equity investors are seeking a total return, which includes both capital appreciation (the increase in the market price of the shares) and dividend income, and they invest in the expectation that the company's management will make decisions that maximise shareholder wealth. Debt investors, on the other hand, are primarily interested in receiving interest income from the debt securities they purchase, and they typically expect to hold those securities until maturity, when the principal is repaid.

4.3 Characteristics of preference shares vs ordinary shares? (2)

Preference shares (preferred stock) have characteristics that differ from ordinary shares. Preference shareholders have a fixed dividend entitlement that takes priority over dividends to ordinary shareholders, and in the event of liquidation they are paid before ordinary shareholders but after debt holders. Unlike ordinary shareholders, preference shareholders typically do not have voting rights; however, they receive a more stable income stream than ordinary shareholders, who may receive variable dividends depending on the company's profitability.

4.4 Why are equity investors owners, and how does this affect their claims? (2)

Equity investors are considered owners because they hold shares in the company, which gives them ownership rights. This ownership provides them with a claim on the company's assets after all liabilities have been settled, meaning they have a residual claim — once the company's debts are paid off, equity shareholders are entitled to any remaining assets or profits. However, unlike debt investors, equity investors do not have guaranteed returns and are exposed to more risk, since their returns depend on the company's performance and market conditions.

4.5 Main advantage for a company issuing equity instead of debt? (2)

The main advantage for a company in issuing equity securities is that it does not incur any obligation to repay the capital raised or to make periodic interest payments, unlike debt securities. This provides the company with more financial flexibility, as there are no fixed repayment schedules, and issuing equity can improve the company's debt-to-equity ratio, making it potentially easier to secure future financing. The trade-off, however, is that issuing equity dilutes existing shareholders' ownership and control.

QUESTION 5 · PRIMARY & SECONDARY MARKETS, IPOS, LISTING [30]

5.1 Main difference between primary and secondary equity markets, and their functions? (5)

The main difference is that the primary market is where a company issues new shares and raises capital, for example through an IPO or a seasoned offering, serving the function of capital formation, while the secondary market is where investors trade existing shares among themselves, with no new capital flowing to the company. The secondary market performs the functions of providing liquidity, price discovery and continuous, fair valuation. The two are closely linked, because investors are willing to buy in the primary market only because the secondary market lets them sell later.

5.2 Explain the IPO process and how it helps a company raise capital. (5)

An IPO is the process by which a company first offers its shares to the public. The company decides to go public, appoints advisers and underwriters, conducts due diligence and publishes a prospectus, sets the offer price (usually through a book-build), obtains regulatory and exchange approval, offers the shares to investors, and then lists and allots them so that trading can begin. It helps the company raise capital by bringing in a large pool of permanent equity, widening the investor base, providing shares that can be used as currency for future acquisitions, and raising the company's profile.

5.3 Advantages and disadvantages of listing on the JSE? (5)

Listing on the JSE has both advantages and disadvantages:

- **Advantages:** access to capital, liquidity for shareholders, enhanced profile and credibility, the use of shares as acquisition currency, easier future fundraising, and a transparent market valuation.
- **Disadvantages:** significant listing and ongoing compliance costs, a heavy disclosure and regulatory burden, some loss of control, short-term market pressure, greater vulnerability to a hostile takeover, and constant public scrutiny.

5.4 Difference between an IPO and a seasoned equity offering (SEO), and the effect of each? (5)

An IPO is a company's first offering of shares to the public and requires a price-setting process, whereas a seasoned equity offering is a further issue of shares by a company that is already listed and does not require price-setting because a market price already exists. An IPO transforms a private company into a public one and raises a large first round of capital, with founders often diluting or partly selling their stakes. An SEO raises additional capital but dilutes existing shareholders unless they take up their rights, and it can send a signal to the market that is sometimes interpreted negatively.

5.5 Role of underwriters in listing equity, and the risks they face? (5)

Underwriters, which are usually investment banks, advise on, value, price and market a share issue, and they guarantee it: in a firm-commitment underwriting they agree to buy any shares not taken up, ensuring that the company raises its target amount. Their main risk is that, if the issue is mispriced or undersubscribed, they are left holding shares that they may have to sell at a loss. They also bear market risk in the period between pricing and sale, and reputational risk if the listing fails.

5.6 Advantages of secondary markets for investors and companies, and their key functions? (5)

Secondary markets benefit investors by providing liquidity through easy entry and exit, price discovery, fair valuation and the ability to diversify. They benefit companies because a liquid, fairly-priced share supports a higher valuation and a lower cost of capital and makes future fundraising easier. Their key functions are providing liquidity, enabling price discovery, allocating capital efficiently and transferring risk between investors.

QUESTION 6 · LISTING REASONS, METHODS, RIGHTS ISSUE, RISKS [15]

GIVEN The JSE offers secure primary & secondary markets; **Company ABC** plans to list on the JSE.

6.1 Reasons for seeking a stock-market listing? (6)

The reasons for seeking a stock-market listing include:

- **Raising capital** for growth and expansion.
- Providing **liquidity and an exit** for existing shareholders.
- Raising the company's **profile and credibility**.
- Obtaining a transparent **market valuation**.
- Using **shares as acquisition currency** and making future fundraising easier.
- Broadening the **investor base** and helping to attract and retain staff through share-incentive schemes.

6.2 Describe two methods of obtaining a listing. (2)

Two methods of obtaining a listing are:

- **Initial Public Offering (offer for subscription/sale):** the company offers shares to the public and raises capital as it lists.
- **Introduction:** already-issued shares are listed without a new offer or any capital being raised, typically where the shareholder base is already sufficiently wide.

6.3 A listed company raises R2bn in equity (rights issue). If you do NOT follow it, what happens to your holding? **(3)**

If you do not follow the rights issue, your shareholding is diluted: your percentage ownership and your voting power fall as new shares are issued. The share price typically adjusts down to a theoretical ex-rights price, which reduces the value of your existing holding. You can sell your nil-paid rights to partly offset the dilution, but if you take no action at all you forfeit that value as well.

6.4 Risks of trading in the equity market? **(4)**

The risks of trading in the equity market include:

- **Market or price risk:** volatility can cause a capital loss, with the price falling below the cost you paid.
- **Liquidity risk:** you may not be able to sell quickly at a fair price.
- **Company-specific (business) risk:** poor performance hurts the share, and dividends are not guaranteed.
- **Systematic / economic risk:** market-wide movements affect the share, with currency risk added where there is foreign exposure.

QUESTION 7 · GORDON GROWTH MODEL [5]

GIVEN Blue-chip telecoms: dividend = **820c**, growth **g = 1%**, required return **r = 8%**.

7.1 Shortcomings of the Gordon Growth model to keep in mind. (5)

Context (the model's output): Value = $D_0(1+g)/(r-g)$ = 820 × 1.01 ÷ (0.08 − 0.01) = 828.2 ÷ 0.07 ≈ **11 831 cents (about R118.31)**.

The investor should, however, keep several shortcomings in mind:

- It assumes a single, **constant growth rate in perpetuity**, which is unrealistic for real companies.
- It is **extremely sensitive to its inputs** — a small change in the required return or the growth rate produces a large change in the calculated value.
- It **breaks down when g ≥ r**, giving a negative or meaningless result.
- It applies only to **mature, stable, dividend-paying companies** — not to high-growth or non-dividend payers.
- It relies heavily on **estimated inputs** and ignores other value drivers and qualitative factors.

QUESTION 8 · MARKET CAPITALISATION — FSR VS NEDBANK [10]

GIVEN FirstRand (FSR) share price **R47.33**; Nedbank (NED) share price **R206.50**. Does NED's higher price mean it is more valuable?

8.1 Describe market capitalisation using these two companies. (4)

Market capitalisation is the total market value of a company's equity, and it is calculated as the current share price multiplied by the number of shares in issue — it is not the share price on its own. The R47.33 and R206.50 quoted for FirstRand and Nedbank are only per-share prices, so to compare the two companies' values you must multiply each price by that company's number of shares outstanding. A company with a lower share price but far more shares in issue can have a much larger market capitalisation.

8.2 Which of the two has the greater equity value? (2)

It is not possible to tell which company is more valuable from the share price alone — the company with the higher market capitalisation has the greater equity value. In practice FirstRand has many more shares in issue than Nedbank, so FirstRand's market capitalisation is larger despite its lower share price, which shows that a higher share price does not mean a more valuable company.

8.3 Describe four benefits of being a listed company. (4)

Four benefits of being a listed company are:

- **Access to capital** for growth and expansion.
- **Liquidity and marketability** of its shares for shareholders.
- **Profile and credibility**, together with a transparent market valuation.
- **Shares as acquisition currency** and easier access to further capital in future.

QUESTION 9 · VALUATION TECHNIQUES BY SECTOR [10]

9.1 Give an example of a listed company in that (telecommunications) sector. (1)

An example of a listed company in the telecommunications sector is MTN Group; other examples include Vodacom Group and Telkom, all listed on the JSE.

9.2 List three valuation techniques, the companies/sectors each suits, and why. (9)

TECHNIQUE	SUITED TO	REASON
Dividend Discount Model (DDM / Gordon Growth)	Mature, stable dividend payers — telecoms, banks, utilities	Regular, predictable dividends make the present value of dividends meaningful.
Discounted Cash Flow (FCFF / FCFE)	Firms with no or irregular dividends but predictable cash flows — industrials, growth firms	Values the intrinsic cash-generating ability directly, even without dividends.
Relative valuation (P/E, EV/EBITDA)	Firms with comparable listed peers and positive earnings — retail, consumer	Gives a quick, market-based benchmark against similar companies.

An asset-based or NAV model also suits asset-heavy firms such as property/REITs and investment holding companies, where value equals total assets minus total liabilities.

QUESTION 10 · DUAL-CLASS SHAREHOLDING CASE STUDY [25]

GIVEN **Naspers** (N shares = 1 vote; unlisted A shares = 1 000 votes, held by management) and **Alphabet/Meta** (multi-class: high-vote insider shares + low/no-vote public shares) — dual-class structures that concentrate control with founders/insiders.

10.1 How do these structures impact corporate governance? (3)

These dual-class structures concentrate control in the hands of insiders and breach the principle of one share, one vote. They weaken the board's accountability to ordinary shareholders and can entrench management. The trade-off is that, while they provide stability and a long-term vision, they also reduce the checks and balances that normally protect minority shareholders.

10.2 Could concentrated insider voting compromise accountability to shareholders? (3)

Yes. Because high-voting shareholders can override ordinary shareholders, they are able to resist shareholder activism and act with fewer checks. Ordinary N or Class A shareholders can never outvote the insiders, no matter how many shares they hold, which materially reduces management's accountability to the broader shareholder base.

10.3 How might these structures affect an investor's decision to buy? (3)

These structures can deter some investors, who see the limited say and the governance risk as reasons to avoid the shares, while others are willing to accept them where management has a strong track record, as with Naspers's transformational investment in Tencent. Ultimately the decision depends on the investor's trust in management and on the price or valuation on offer.

10.4 **Do investors perceive higher risk from diminished voting power, and how is it reflected in price/valuation?** (3)

Yes. Diminished voting power is viewed as a governance risk, and this is often reflected in a governance discount, where the low- or no-vote shares — such as Naspers's N shares or Alphabet's Class C shares — tend to trade at a discount to the high-vote shares. Investors demand a higher required return to compensate for the weaker rights, and this translates into a lower valuation.

10.5 **Would the companies make different strategic decisions if voting power were more equal?** (2)

Quite possibly. With votes distributed more equally, management would face greater short-term shareholder pressure and might make less bold, less long-term decisions. Transformational bets such as Naspers's early investment in Tencent might not have been made under that kind of pressure.

10.6 **How do these structures serve as a defence against hostile takeovers?** (2)

Because the insiders' concentrated votes mean that an acquirer cannot gain control simply by buying up the publicly traded shares, the structure acts as an effective shield against hostile takeovers.

10.7 **Other mechanisms to maintain control without multi-class shares?** (2)

Yes — companies can use mechanisms such as poison pills, staggered or classified boards and supermajority voting provisions, as well as golden shares, shareholder agreements or a pyramid/holding-company structure, to maintain control without resorting to multi-class shares.

10.8 How do these structures affect market perception of transparency & governance? (2)

They can signal weaker governance and transparency, which is a reputational concern and has led some stock-market indices to exclude dual-class companies. However, consistently strong performance can offset this concern and keep investors comfortable with the structure.

10.9 How sustainable are these structures in the long run? (3)

Their sustainability is mixed. They can persist for as long as the founders continue to deliver strong performance and retain their high-vote shares, but pressure builds over time through shareholder activism, index exclusion and succession when founders eventually exit. Sunset clauses that automatically convert the shares to one-share-one-vote are increasingly common, and the structures may prove unsustainable if performance falters.

10.10 Could these companies alter their structures in future? (2)

Yes — a company might alter its structure if a founder exits, dies or dilutes below a key threshold, if investor, regulatory or index pressure becomes strong enough, if a sunset clause triggers a conversion to one-share-one-vote, or if a large capital requirement forces it to issue broader equity.

SUBMISSION REMINDER Write your final answers in your **own words**, number them to match the workbook, and put your name, surname & ID on every page. Pass mark = **65%**.